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Enhanced Monocular Depth Estimation Using ResNet Backbone and Multiscale Feature Decoding for Precise Object Distance Measurement

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ABSTRACT

Monocular depth estimation is a key capability for intelligent transportation systems and Internet of Things (IoT)-enabled cyber-physical environments, where scalable, low-cost, and real-time perception is essential. Traditional approaches rely on stereo vision or LiDAR, increasing deployment cost and limiting large-scale adoption. This paper presents a deep learning-based framework that produces high-fidelity depth maps for long-range scenes up to 250 m using the D-Far250 dataset. The architecture employs a ResNet-based encoder–decoder with multiscale supervision to improve depth consistency across near and far-range regions. To enable practical perception for autonomous systems, the depth module is integrated with the YOLOv8 object detector to estimate object-level distances directly from monocular images. Experimental evaluation on both real and synthetic datasets demonstrates strong accuracy and generalization. The model achieves a (δ_1) accuracy of 94.98% on D-Far250 and 90.85% on the KITTI. The framework operates at over 200 FPS on an NVIDIA RTX 4070 GPU, confirming suitability for latency-critical IoT applications. By combining accurate monocular depth prediction with detection-driven distance inference, the proposed system enables scalable, LiDAR-free perception for collision avoidance, autonomous navigation, unmanned aerial vehicle (UAV) operations, and smart surveillance in connected IoT ecosystems and advanced urban mobility services worldwide and beyond.

1 | Introduction

The widespread adoption of the Internet of Things (IoT) is transforming cyber-physical mobility systems by enabling data-driven autonomy in transportation, robotics, and large-scale smart infrastructure. Among the sensing modalities used in these systems, cameras remain the most cost-efficient, lightweight, and easily deployable option compared to LiDAR or stereo rigs, which suffer from high hardware costs, calibration constraints, and scalability limitations. As a result, monocular depth estimation has emerged as a compelling alternative for enabling low-cost

3D perception within IoT-enabled cyber-physical environments. However, achieving reliable long-range and metrically accurate depth estimation from a single RGB image remains a fundamental challenge, especially in safety-critical applications such as collision avoidance, lane-level navigation, and connected vehicle coordination [1, 2].

Traditional monocular depth estimation methods [3–5], whether based on supervised regression, self-supervised view synthesis, or more recent transformer-based architectures, typically produce low-precision depth maps (8–10 bit) and are limited to

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short-range outdoor perception (up to 80 m), as constrained by datasets such as KITTI [6]. These limitations hinder their suitability for long-range reasoning in highway or high-speed mobility environments, where vehicles and infrastructure must anticipate objects well beyond 100 m. Moreover, many state-of-the-art models predict relative depth rather than absolute metric depth, necessitating additional post-hoc scale alignment, which can introduce instability in real-world deployments.

To overcome these constraints, we propose a unified monocular depth estimation framework that produces high-fidelity 16-bit metric depth maps and supports both short-range (0–80 m, KITTI) and extended long-range (up to 250 m) perception. The model employs a lightweight ResNet-based encoder–decoder with multiscale supervision to preserve geometric consistency and depth granularity across varying spatial scales. To couple depth understanding with semantic awareness, we integrate the YOLOv8 model for object detection, enabling direct object-wise distance estimation from the predicted depth maps without relying on LiDAR or stereo cues. This combination enables the system to infer accurate distances to vehicles, cyclists, pedestrians, and infrastructure elements from monocular imagery alone, making it well-suited for IoT-driven intelligent transportation environments.

Real-world depth datasets rarely exceed 80 m, creating a persistent gap in evaluating long-range monocular perception. To address this, we introduce D-Far250, a synthetic dataset designed explicitly for long-range depth estimation and object-distance benchmarking up to 250 m. D-Far250 provides dense 16-bit depth annotations and diverse road scenes, enabling systematic evaluation of extended-range performance. Using the same architecture and training strategy across KITTI and D-Far250, we demonstrate that the proposed model maintains stable metric fidelity across both domains.

The main contributions of this work are as follows:

- We propose a lightweight, high-fidelity monocular depth estimation framework that generates 16-bit metric depth maps for ranges up to 80 m (KITTI) and 250 m (D-Far250), enabling fine-grained depth representation suitable for IoT-based intelligent transportation systems.
- We introduce D-Far250, a new synthetic dataset tailored for long-range monocular depth estimation and distance-aware object evaluation, supporting metric ground truth up to 250 m.
- We integrate YOLOv8 with depth prediction to estimate object-wise distances, forming a unified pipeline for low-cost, LiDAR-free spatial perception.

The remainder of this paper is organized as follows: Section 2 reviews relevant literature on monocular depth estimation and detection-guided perception. Section 3 details the proposed architecture and system workflow. Section 4 presents quantitative and qualitative evaluations. Section 5 discusses key findings and limitations. Finally, Section 6 concludes the paper and outlines future research directions.

2 | Related Work

Monocular depth estimation has garnered significant attention in recent years due to its potential to replace expensive, power-hungry depth sensors, such as LiDAR. We review the literature across five major thematic categories to provide context for our work.

2.1 | Supervised, Unsupervised Learning, and Self-Supervised Monocular Depth Estimation

Early work in monocular depth estimation focused on learning from ground-truth depth labels in a fully supervised manner [7]. Eigen et al. [1] pioneered the use of multiscale convolutional neural networks for depth estimation. Laina et al. [3] leveraged fully convolutional residual networks (FCRN) and introduced a reverse Huber loss to improve convergence. Fu et al. [4] proposed a deep ordinal regression network, interpreting depth as a classification task. Lee et al. [8] applied attention mechanisms for context aggregation, while Hu et al. [9] emphasized feature fusion via self-attention and scale-aware modules. Several datasets such as NYU-Depth V2 (New York University Depth Dataset V2) [10], KITTI [2], and Make3D [11] have facilitated benchmarking these methods. However, supervised methods rely heavily on dense ground truth, which is costly and often unavailable in outdoor datasets such as KITTI. Our method addresses this limitation by standardizing supervision in a 16-bit metric-depth format and introducing D-Far250 to enable dense, long-range ground truth for scalable training and evaluation beyond short-range real-world benchmarks.

To mitigate the need for ground-truth depth, self-supervised methods have emerged. Zhou et al. [12] proposed unsupervised learning using view synthesis between consecutive frames. Godard et al. [13, 14] introduced left–right photometric loss for stereo supervision. Monodepth2 [14] further refined the pipeline by incorporating pose networks and auto-masking strategies.

Unsupervised methods such as [15] leverage geometric consistency using 3D point clouds. Later, Shu et al. [16] used feature matching to improve unsupervised monocular depth estimation. Despite their label-free advantage, most of these methods output normalized floating-point depth, lacking high-resolution scalability. In contrast, our method outputs 16-bit depth maps aligned with real-world metrics.

Attention mechanisms and transformer architectures have recently shown strong performance. Ranftl et al. [17] introduced MiDaS, a multidataset model using ViT backbones for robust cross-dataset generalization. Bhat et al. [18] proposed AdaBins, using transformer layers to bin depth ranges dynamically. Yuan et al. [19] improved transformer generalization with monocular cues. While powerful, transformer-based models require substantial compute and often prioritize computational efficiency over real-time performance. Our system balances accuracy and runtime by employing a ResNet-based encoder and a modified decoder with a lightweight inference strategy.

Joint learning of depth with related tasks such as segmentation, normal prediction, or flow estimation has shown promising results. Kendall et al. [20] proposed task uncertainty-based loss

weighting in multitask setups. Xu et al. [21] integrated depth, normals, and semantic cues into a shared representation. Multimodal inputs, such as stereo pairs, infrared imagery, or event data, have also been incorporated [22, 23]. However, these methods require specialized sensors or dual-view setups, whereas our method achieves distance-aware detection using only monocular RGB and a unified deep architecture.

2.2 | Object Detection + Depth Fusion for Distance Estimation

Few works have combined object detection with depth for distance inference. Chen et al. [24, 25] performed 3D bounding box regression using monocular depth priors. Fang et al. [26] fused semantic detection with attention-based depth refinement. YOLO-based distance estimation has been explored for autonomous driving [27], primarily using stereo or pseudo-LiDAR data. Our approach advances this domain by detecting objects with YOLOv8 and estimating their precise distances using our predicted 16-bit depth maps, enabling accurate, lightweight spatial awareness in a monocular setup.

2.3 | Synthetic Datasets for Long-Range Depth Estimation

While extensive research has been conducted on monocular depth estimation using datasets such as KITTI, NYU-Depth V2 [10], and Cityscapes [28], these datasets are typically limited to short- or mid-range perception (up to 80 m). However, safety-critical applications in autonomous driving, particularly on highways or open terrains, require perception systems to reason about objects located at distances beyond 100 m.

Existing works, such as [29, 30] and [18], aim to generalize but still operate within traditional dataset limits. To bridge this gap, we introduce a synthetic dataset, D-Far250, that spans up to 250 m. The dataset is explicitly designed for long-range monocular depth estimation and object-wise distance inference. It contains synthetically rendered scenes with pixel-level depth annotations and object bounding boxes at extended distances, replicating realistic urban and highway driving scenarios. Our model, trained with the same architecture and loss functions used for KITTI, demonstrates strong adaptability and precision on D-Far250, validating the robustness of our depth estimation framework across diverse spatial scales.

2.4 | State of the Art and Emerging Trends

Recent progress in monocular depth estimation centers on three trends. First, large-scale pretrained models such as DPT/MiDaS, ZoeDepth, and Depth-Anything use millions of pseudo-depth samples to improve cross-dataset generalization, typically predicting high-quality relative depth that is later converted to metric form through lightweight alignment modules. Second, new methods focus on improving metric scale recovery by incorporating camera intrinsics, geometric cues, or dedicated scale-prediction heads to produce metrically consistent depth suitable for robotics and advanced driver assistance systems (ADAS). Third, although

transformer backbones offer strong accuracy, deployment constraints are driving interest in compact CNN/ViT-hybrid decoders, model distillation, and low-precision INT8/FP16 inference for real-time edge operation. Despite these advances, long-range depth estimation remains underexplored because most benchmarks are limited to roughly 80 m and lack high-bit-depth annotations. Our work addresses this gap by producing 16-bit metric depth and evaluating on both KITTI (0–80 m) and D-Far250 (0–250 m), enabling reliable long-range and object-wise distance estimation.

Table 1 provides a comparative summary of widely cited monocular depth estimation methods, highlighting their key innovations, datasets used, and existing limitations. This contextualizes the positioning of our proposed method with respect to both performance and practical deployment considerations.

3 | Methods

This study presents a unified monocular perception pipeline that performs high-fidelity depth estimation and object-wise distance inference from a single RGB image (Figure 1). The framework integrates three components into a cohesive system: a lightweight encoder-decoder network trained to generate 16-bit metric depth maps scaled to the appropriate range for each dataset (0–80 m for KITTI and 0–250 m for D-Far250), a YOLOv8 object detection model fine-tuned for the same domains, and a post-processing module that fuses detected bounding boxes with corresponding pixel-level depth values to compute real-world distances. Unlike conventional monocular methods that produce low-precision or short-range depth estimates, the proposed design incorporates hybrid multiscale supervision and long-range refinement to maintain metric consistency over extended distances. Furthermore, the introduction of the D-Far250 synthetic benchmark and the use of a 16-bit depth format enable evaluation beyond the 80 m limitation of KITTI, providing an end-to-end solution for both short- and long-range monocular depth estimation and object localization.

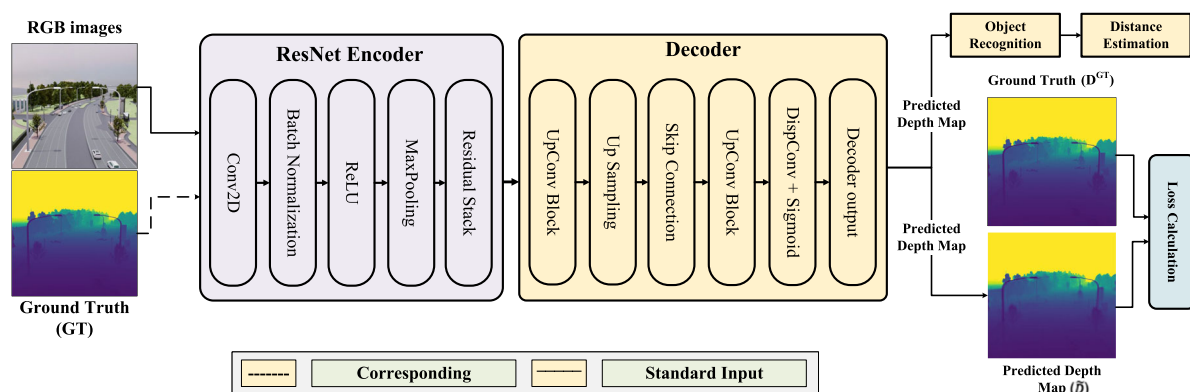
3.1 | Datasets

The KITTI depth completion and prediction dataset [2] provides real-world urban driving scenes with ground-truth depth collected using LiDAR. Depth annotations are available up to 80 m, making it the most widely used benchmark for supervised monocular depth estimation. KITTI's sparse ground-truth maps are particularly valuable for evaluating short- to mid-range perception in realistic driving environments. The KITTI dataset is a publicly available benchmark dataset widely used in autonomous driving research. Its use in this study complies with the dataset's official licensing terms and does not involve any personal or sensitive information.

To overcome KITTI's 80 m limitation, we introduce a synthetic dataset, D-Far250, designed to evaluate long-range monocular depth estimation and object-aware distance prediction. D-Far250 simulates highway driving conditions with dense ground-truth annotations for objects up to 250 m away. All ground-truth depth maps are stored in 16-bit format, ensuring

TABLE 1 | Comparative analysis of state-of-the-art monocular depth estimation methods.

Author(s)	Technique	Dataset(s)	Key method	Limitations
Eigen et al. [1]	Multiscale CNN	NYU, KITTI	Supervised regression with coarse-to-fine prediction	Lacks object-aware distance estimation; no 16-bit support
Laina et al. [3]	FCRN + ResNet	NYU V2	Reverse Huber Loss, fully convolutional	Fixed output scale; no object-level inference
Fu et al. [4]	Ordinal Regression	KITTI	Depth as classification with discretization	Classification-based depth less precise; lacks detection
Godard et al. [14]	Monodepth2	KITTI	Self-supervised stereo matching	Output not scaled for real-world distances; lacks detection
Bhat et al. [18]	AdaBins	NYU, KITTI	Transformer with adaptive binning	Slower inference; lacks detection, distance fusion
Yuan et al. [19]	MDEFormer	KITTI, NYU	Transformer-based monocular depth	High complexity; lacks 16-bit output
Ranftl et al. [17]	DPT/MiDaS	Multiple	Multidataset ViT; robust cross-domain	Heavy backbone; metric scale requires alignment
Bhat et al. [31]	Scale-aware depth head	KITTI, NYU	Relative-to-metric conversion; strong generalization	Overhead in inference; limited >80 m validation
Wang et al. [32]	Large-scale pretraining	Multiple	Broad robustness, transferable encoders	Primarily relative depth; metric scale needs calibration
Zhang et al. [33]	Camera-aware metric head	Outdoor sets	Direct metric predictions, improved scale recovery	Calibration-sensitive; limited long-range study
LiWei et al. [34]	Efficient encoders	KITTI, NYU	Low latency inference	Accuracy gap vs ViTs; little long-range evaluation

**FIGURE 1** | Overview of the proposed system integrating 16-bit monocular depth estimation with YOLOv8 object detection for accurate object-wise distance inference.

high-fidelity metric scaling suitable for safety-critical perception tasks. This dataset allows us to benchmark model performance beyond conventional ranges, validating scalability in extended environments. Since D-Far250 is synthetically generated using a rendering engine, it does not contain any personal or human subject data, ensuring compliance with standard research ethics and data protection regulations.

The D-Far250 dataset was generated using a Blender-based synthetic rendering engine configured with calibrated camera intrinsics (focal length $f = 721.5$ px, principal point ($c_x = 609.6$, $c_y = 172.8$)), matching the KITTI calibration settings. Scenes were rendered at a native resolution of 512×512 with a

forward-translation velocity of 10–12 m/s and a 60-m baseline camera trajectory to replicate real driving motion. The rendered 32-bit depth was subsequently quantized and stored as 16-bit PNG to ensure consistent training and evaluation in a compact, high-fidelity format. Illumination was modeled using physically based HDR lighting with a light intensity of 120,000–140,000 Lx and an ambient occlusion radius of 0.5–0.8 m to maintain realistic outdoor visibility. Scene geometry includes variable road widths (3.2–3.8 m), building and tree heights ranging from 8 to 18 m, and vegetation densities that approximate transitions between urban and suburban environments. Realism was validated by ensuring that horizon-line projection, vanishing-point geometry, and long-range depth gradients

matched KITTI reference sequences within a $\pm 3\%$ deviation in pixel-depth slope error.

The D-Far250 dataset was generated using the Blender rendering engine for its flexibility, physically based rendering capabilities, and widespread adoption in computer vision research for synthetic data generation. Blender provides precise control over scene geometry, lighting conditions, camera parameters, and material properties, making it suitable for creating large-scale depth datasets with consistent ground truth. During dataset generation, vehicle models were assigned physically based material parameters, including diffuse reflectance, roughness, and specular properties, to approximate real-world surface characteristics. Environmental lighting and road textures were also configured to simulate typical outdoor driving conditions. These configurations help produce visually realistic scenes while maintaining accurate depth annotations required for long-range distance estimation experiments.

3.2 | Preprocessing

Before training or inference, each RGB image and its corresponding depth map undergo preprocessing to ensure numerical stability, format consistency, and alignment for paired training.

RGB Normalization: Before training, all RGB image pixels are normalized to the range $[0,1]$ and standardized using the ImageNet mean and standard deviation. This step ensures numerical stability by preventing large input magnitudes that could lead to exploding or vanishing gradients during backpropagation, and (ii) it aligns the input distribution with that of pre-trained ResNet, which is optimized under normalized conditions. Let the raw RGB input image be $I_{\text{raw}} \in \mathbb{R}^{3 \times H \times W}$. It is normalized using ImageNet mean and standard deviation

$$\mu = [0.485, 0.456, 0.406], \quad \sigma = [0.229, 0.224, 0.225]$$

$$I_{\text{norm}} = \frac{I_{\text{raw}} - \mu}{\sigma}$$

Depth Map Scaling: Ground truth depth maps are converted to a unified 16-bit PNG representation with pixel intensities in the range $[0, 65535]$ to ensure consistent scaling across KITTI and D-Far250. To convert them to real-world metric depth $D(x,y)$, we apply dataset-specific scaling

$$D(x,y) = \frac{D_{\text{raw}}(x,y)}{65535} \cdot D_{\text{max}}$$

$$\text{where } D_{\text{max}} = \begin{cases} 80 \text{ m,} & \text{KITTI dataset} \\ 250 \text{ m,} & \text{D - Far250 dataset} \end{cases}$$

This produces a normalized depth map in which each pixel represents the actual distance to the camera. All training and evaluation steps are conducted in a metric space. Both datasets use a unified augmentation pipeline: random horizontal flipping (50% probability), color jittering, and resolution standardization (resizing using bilinear for RGB and nearest-neighbor for depth). These augmentations preserve geometric alignment between RGB and depth modalities while improving generalization.

3.3 | Encoder-Decoder Architecture

The proposed monocular depth estimation network follows an encoder-decoder architecture that extracts hierarchical visual features and progressively reconstructs dense depth predictions. Let the input RGB image be denoted as $I \in \mathbb{R}^{3 \times H \times W}$. The encoder extracts multiscale feature representations, while the decoder progressively upsamples and fuses these representations through skip connections to generate the final depth map.

3.3.1 | Encoder

The encoder is implemented using a ResNet backbone, which transforms the input image into hierarchical feature maps at multiple spatial resolutions. The initial convolutional layer applies a 7×7 convolution with a stride of 2, followed by batch normalization and a rectified linear unit (ReLU) activation. Formally, the initial feature extraction is defined in Equation (1), where $\sigma(\cdot)$ denotes the ReLU activation function.

$$\mathbf{F}_0 = \sigma(\text{BN}(\text{Conv}_{7 \times 7}^s(I))) \quad (1)$$

A max-pooling operation reduces the spatial resolution $\mathbf{F}_1 = \text{MaxPool}(\mathbf{F}_0)$. The encoder then applies four residual stages that progressively increase feature abstraction while reducing spatial resolution. The hierarchical residual feature extraction is described in Equation (2), where $\mathcal{R}_k(\cdot)$ represents the residual block sequence at stage k . Each stage produces a feature map $\mathbf{F}_k \in \mathbb{R}^{C_k \times H_k \times W_k}$.

$$\mathbf{F}_k = \mathcal{R}_k(\mathbf{F}_{k-1}), \quad k \in \{1, 2, 3, 4\} \quad (2)$$

In the ResNet-18 configuration used in this work, the channel dimensions are

$$[C_0, C_1, C_2, C_3, C_4] = [64, 64, 128, 256, 512].$$

Thus the encoder produces a set of hierarchical feature representations $\mathcal{F} = \{\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \mathbf{F}_4\}$, where spatial resolutions scaling follows Equation (3).

$$(H_k, W_k) = \left(\frac{H}{2^{k+1}}, \frac{W}{2^{k+1}} \right) \quad (3)$$

These feature maps capture low-level texture information at shallow layers and high-level semantic representations at deeper layers.

3.3.2 | Decoder

The decoder reconstructs high-resolution depth predictions by progressively upsampling the deepest encoder feature map while integrating skip connections from earlier encoder stages. Let the deepest encoder feature be $\mathbf{X}_4 = \mathbf{F}_4$. At each decoder stage i , the feature map is first processed by a convolution block. The decoder convolution operation is defined in Equation (4), where $\phi(\cdot)$ denotes the exponential linear unit (ELU) activation function.

$$\mathbf{Z}_i = \phi(\text{Conv}_{3 \times 3}(\mathbf{X}_i)) \quad (4)$$

The feature map is then upsampled by a factor of two using bilinear interpolation. The upsampling operation is expressed in Equation (5), where $\text{Up}(\cdot)$ denotes bilinear upsampling.

$$\tilde{\mathbf{Z}}_i = \text{Up}(\mathbf{Z}_i) \quad (5)$$

If skip connections are enabled, the corresponding encoder feature map is resized to the same spatial resolution and concatenated with the upsampled decoder feature. The skip-connection fusion mechanism is defined in Equation (6).

$$\mathbf{S}_i = \text{Concat}(\tilde{\mathbf{Z}}_i, \mathbf{F}_{i-1}) \quad (6)$$

The fused representation is then refined using another convolutional block, and the fused decoder representation is computed according to Equation (7).

$$\mathbf{X}_{i-1} = \phi(\text{Conv}_{3 \times 3}(\mathbf{S}_i)) \quad (7)$$

This hierarchical reconstruction continues for $i = 4, 3, 2, 1$, progressively recovering spatial resolution.

At selected decoder scales, a 3×3 convolution layer predicts a single-channel normalized depth map using Equation (8).

$$\mathbf{D}_s = \sigma(\text{Conv}_{3 \times 3}(\mathbf{X}_s)) \quad (8)$$

where $\sigma(\cdot)$ denotes the sigmoid activation and s indicates the prediction scale. The final prediction is taken from the highest resolution decoder output $\hat{\mathbf{D}} = \mathbf{D}_0$, with values normalized to the range $[0,1]$.

If a prior depth estimate $\mathbf{P}$ is available, it is resized to match the predicted depth resolution and used to refine the prediction.

3.4 | Depth Prediction and Metric Conversion

The decoder produces a single-channel *normalized depth* prediction in the range $[0,1]$. Let the encoder-decoder network be denoted by $f_\theta(\cdot)$ with parameters θ . Given an input RGB image $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$, the network outputs as Equation (9).

$$\hat{\mathbf{Z}}_{\text{norm}} = f_\theta(\mathbf{I}), \quad \hat{\mathbf{Z}}_{\text{norm}} \in [0, 1]^{H \times W} \quad (9)$$

The normalized prediction is produced by applying a sigmoid activation at the final prediction layer to ensure bounded outputs.

3.4.1 | Ground-Truth Normalization for Supervision

Let $\mathbf{Z}_{\text{metric}} \in \mathbb{R}^{H \times W}$ denote the ground-truth depth map in meters. To stabilize training and keep gradients numerically well-scaled, supervision is performed in normalized space through Equation (10).

$$\mathbf{Z}_{\text{norm}} = \frac{\mathbf{Z}_{\text{metric}}}{Z_{\text{max}}} \quad (10)$$

where Z_{max} is the dataset-specific maximum depth range

$$Z_{\text{max}} = \begin{cases} 80, & \text{KITTI} \\ 250, & \text{D - Far250} \end{cases} \quad (11)$$

During optimization, the loss is computed between $\hat{\mathbf{Z}}_{\text{norm}}$ and $\mathbf{Z}_{\text{norm}}$ over valid pixels.

3.4.2 | Metric Depth Reconstruction (Inference)

At inference time, the normalized depth prediction is converted back to metric depth (meters) through Equation (12).

$$\hat{\mathbf{Z}}_{\text{metric}} = Z_{\text{max}} \hat{\mathbf{Z}}_{\text{norm}}, \quad \hat{\mathbf{Z}}_{\text{metric}} \in [0, Z_{\text{max}}]^{H \times W} \quad (12)$$

3.4.3 | 16-Bit Depth Map Encoding

For storage and interoperability, the metric depth map is encoded as a 16-bit PNG image. The 16-bit depth encoding is defined in Equation (13).

$$\mathbf{Z}_{16} = \left\lfloor \frac{\hat{\mathbf{Z}}_{\text{metric}}}{Z_{\text{max}}} \cdot 65535 \right\rfloor, \quad \mathbf{Z}_{16} \in \{0, \dots, 65535\}^{H \times W} \quad (13)$$

This encoding preserves fine-grained depth precision while maintaining a compact image format suitable for real-time pipelines and downstream object-wise distance estimation.

3.5 | Loss Function and Optimization

To train the proposed depth estimation network, the predicted depth map is supervised using ground-truth depth measurements. Let $\hat{\mathbf{Z}}$ denote the predicted normalized depth map and $\mathbf{Z}$ represent the ground truth depth map. The training objective minimizes the discrepancy between these two maps over valid pixels.

3.5.1 | Mean Squared Error Loss

The mean squared error (MSE) loss measures the average squared difference between the predicted and ground-truth depths. It is defined as

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (\hat{Z}_i - Z_i)^2 \quad (14)$$

where N denotes the total number of valid pixels in the depth map and $\hat{Z}_i$ and Z_i correspond to the predicted and ground-truth depth values at pixel i .

3.5.2 | Huber Loss

To improve robustness against outliers and noisy depth values, the training framework primarily employs the Huber loss. The Huber loss combines the properties of $L1$ and $L2$ losses and is defined as

$$\mathcal{L}_{\text{Huber}} = \frac{1}{N} \sum_{i=1}^N \ell_\delta(\hat{Z}_i - Z_i) \quad (15)$$

where the function $\ell_\delta(\cdot)$ is defined as

$$\ell_\delta(x) = \begin{cases} \frac{1}{2}x^2, & |x| \leq \delta \\ \delta \left(|x| - \frac{1}{2}\delta \right), & |x| > \delta \end{cases} \quad (16)$$

and δ is a threshold parameter controlling the transition between quadratic and linear error behavior.

In the proposed implementation, the Huber loss is used as the primary training objective because it stabilizes optimization in the presence of noisy depth measurements.

3.5.3 | Final Training Objective

The network is trained using the Huber loss computed between the predicted normalized depth and the normalized ground-truth depth over valid pixels. Therefore, the final objective is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Huber}} \quad (17)$$

$$\begin{aligned} \text{RMSE} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{Z}_i - Z_i)^2}, & \text{MAE} &= \frac{1}{N} \sum_{i=1}^N |\hat{Z}_i - Z_i|, & \text{AbsRel} &= \frac{1}{N} \sum_{i=1}^N \frac{|\hat{Z}_i - Z_i|}{Z_i}, \\ \text{SqRel} &= \frac{1}{N} \sum_{i=1}^N \frac{(\hat{Z}_i - Z_i)^2}{Z_i}, & \text{RMSE}_{\log} &= \sqrt{\frac{1}{N} \sum_{i=1}^N (\log \hat{Z}_i - \log Z_i)^2}, & r_i &= \max\left(\frac{Z_i}{\hat{Z}_i}, \frac{\hat{Z}_i}{Z_i}\right), \\ \delta_k &= \frac{1}{N} \sum_{i=1}^N \mathbb{I}(r_i < 1.25^k), & k &\in \{1, 2, 3\}. \end{aligned} \quad (18)$$

3.5.5 | Optimization

The network parameters θ are optimized using the Adam optimizer. At each training iteration, the parameters are updated according to

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} \mathcal{L}_{\text{total}}, \quad (19)$$

where η denotes the learning rate and $\nabla_{\theta} \mathcal{L}_{\text{total}}$ represents the gradient of the loss with respect to the network parameters.

Early stopping is employed during training to prevent overfitting. The model achieving the lowest validation AbsRel is selected as the final model.

3.6 | YOLOv8 Object Detection

To complement our depth estimation pipeline with semantic scene understanding, we utilize object detection via the v8 model. The detector is trained using both real and synthetic datasets: the KITTI object detection benchmark for short-range scenes, and the newly developed D-Far250 dataset for long-range scenarios.

YOLOv8 is a one-stage detector that predicts bounding boxes and class probabilities over a dense spatial grid. Let the input image be $I \in \mathbb{R}^{3 \times H \times W}$. The detection head outputs a tensor of shape $S \times S \times (B \cdot 5 + C)$, where S is the grid size, B is the number of bounding boxes per grid cell. Each box predicts 5 values: $(x, y, w, h, \text{conf})$, C is the number of object classes.

The predicted bounding box coordinates are parameterized as

$$\hat{x} = \sigma(t_x) + c_x, \quad \hat{y} = \sigma(t_y) + c_y$$

3.5.4 | Evaluation Metrics

To evaluate the accuracy of the predicted depth maps, several standard metrics are used, including error-based and threshold-based measures. Error metrics quantify the difference between predicted depth $\hat{Z}_i$ and ground-truth depth Z_i over N valid pixels, while threshold metrics measure the proportion of predictions within specific accuracy bounds. The rootmean-square error (RMSE), mean absolute error (MAE), absolute relative error (AbsRel), squared relative error (SqRel), and logarithmic RMSE (RMSE_{log}) are defined together with the threshold accuracy metrics δ_k as follows

$$\hat{w} = s \cdot \sigma(t_w), \quad \hat{h} = s \cdot \sigma(t_h)$$

where t_x, t_y, t_w , and t_h denote the predicted offsets, $\sigma(\cdot)$ represents the sigmoid activation function, and s denotes the stride of the detection feature map. In YOLOv8, bounding boxes are predicted using an anchor-free formulation where object dimensions are directly estimated relative to the feature map resolution.

After training on both datasets, the detector generalizes object localization across varying ranges. At inference time, for each RGB frame, YOLOv8 provides B_i : a set of bounding boxes $\{b_1, b_2, \dots, b_n\}$, C_i : corresponding class labels $\{c_1, c_2, \dots, c_n\}$, S_i confidence scores. Nonmaximum suppression is applied using an Intersection-over-Union (IoU) threshold to suppress overlapping predictions

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

The final detection boxes are fused with predicted depth maps to enable per-object distance estimation, even for objects up to 250 m away in D-Far250. This modular setup enables robust, short-, and long-range semantic awareness from a single monocular RGB image.

3.7 | Distance Estimation

The predicted depth map provides per-pixel distance information for the entire scene. To estimate the distances to detected objects, the depth map is combined with bounding-box detections from an object detector. Let the object detector produce a set of bounding boxes $\mathcal{B} = \{B_1, B_2, \dots, B_k\}$, where each bounding box is defined as $B_k = (x_k, y_k, w_k, h_k)$ with (x_k, y_k) representing the

center of the bounding box and (w_k, h_k) representing its width and height.

The bounding box coordinates are converted into pixel region boundaries as

$$x_k^{\min} = x_k - \frac{w_k}{2}, \quad x_k^{\max} = x_k + \frac{w_k}{2} \quad (20)$$

$$y_k^{\min} = y_k - \frac{h_k}{2}, \quad y_k^{\max} = y_k + \frac{h_k}{2} \quad (21)$$

The corresponding depth region extracted from the predicted depth map $\hat{\mathbf{Z}}$ is therefore

$$\mathbf{Z}_k = \hat{\mathbf{Z}}[x_k^{\min} : x_k^{\max}, y_k^{\min} : y_k^{\max}] \quad (22)$$

Since monocular depth predictions may be noisy near object boundaries, a robust statistical aggregation strategy is used to estimate object distances.

Let $\mathcal{P}_k$ denote the set of valid depth pixels inside the bounding box B_k . The estimated object distance is computed as the median depth value $D_k = \text{median}(\mathcal{P}_k)$. The median statistic is chosen instead of the mean because it is less sensitive to outliers caused by occlusions, reflective surfaces, or boundary artifacts.

To further improve robustness, invalid or extreme depth values are discarded before aggregation. Let $\mathcal{P}_k^{\text{valid}}$ denote the filtered set of pixels satisfying $Z_{\min} < z_i < Z_{\max}$. The final object distance estimate becomes

$$D_k = \text{median}(\mathcal{P}_k^{\text{valid}}) \quad (23)$$

The final output of the system for each detected object consists of the tuple $\mathcal{O}_k = (c_k, B_k, D_k)$, where c_k is the predicted object class, B_k is the bounding box location, and D_k is the estimated object distance in meters. Thus, the proposed framework produces both semantic object detection and metric distance estimation for all detected objects in the scene.

ALGORITHM 1 | Robust Object-Wise Distance Estimation from Predicted Metric Depth.

Require: Predicted metric depth map $\hat{\mathbf{Z}} \in \mathbb{R}^{H \times W}$ (meters)

Bounding box $B_j = (x_j, y_j, w_j, h_j)$ for object j (center format)

Minimum valid-pixel threshold τ

Maximum depth-variance threshold ε

Valid depth bounds $(Z_{\min}, Z_{\max})$

Ensure: Estimated distance D_j for object j (meters) or *rejection*

1: Convert box to corners:

$$x^{\min} \leftarrow \lfloor x_j - \frac{w_j}{2} \rfloor, \quad x^{\max} \leftarrow \lceil x_j + \frac{w_j}{2} \rceil$$

$$y^{\min} \leftarrow \lfloor y_j - \frac{h_j}{2} \rfloor, \quad y^{\max} \leftarrow \lceil y_j + \frac{h_j}{2} \rceil$$

2: Clip to image bounds:

$$x^{\min} \leftarrow \max(0, x^{\min}), \quad x^{\max} \leftarrow \min(W-1, x^{\max})$$

$$y^{\min} \leftarrow \max(0, y^{\min}), \quad y^{\max} \leftarrow \min(H-1, y^{\max})$$

3: Extract region $\mathbf{Z}_j \leftarrow \hat{\mathbf{Z}}[y^{\min} : y^{\max}, x^{\min} : x^{\max}]$

4: Form valid pixel set:

$$\mathcal{P}_j \leftarrow \{z \in \mathbf{Z}_j \mid z \neq 0 \wedge Z_{\min} < z < Z_{\max}\}$$

5: **if** $|\mathcal{P}_j| < \tau$ **then**

6: **return** *Reject: insufficient valid depth samples*

7: **end if**

8: Compute dispersion $\sigma_j \leftarrow \text{std}(\mathcal{P}_j)$

9: **if** $\sigma_j > \varepsilon$ **then**

10: **return** *Reject: unstable depth region (high variance)*

11: **end if**

12: Estimate object distance (robust):

$$D_j \leftarrow \text{median}(\mathcal{P}_j)$$

13: **return** D_j

4 | Results

This section presents the quantitative and qualitative evaluation of the proposed monocular perception framework on the KITTI and D-Far250 datasets. We assess performance across three dimensions: depth estimation accuracy, long-range generalization, and object-wise distance prediction.

4.1 | Dataset and Experimental Setup

Experiments were performed on two datasets: the KITTI depth prediction benchmark and the proposed D-Far250 synthetic dataset. KITTI provides RGB images paired with sparse LiDAR depth, with depth values valid up to 80 m. D-Far250 extends this range by offering dense 16-bit depth annotations up to 250 m along with synthetic object labels, enabling evaluation beyond the limitations of real-world sensors. For KITTI, 80% of the scenes were used for training and 20% for validation, while D-Far250 followed the same split to maintain consistency. All RGB frames from both datasets were resized to 690×192 for KITTI and 512×512 for D-Far250, and normalized using ImageNet statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Depth ground-truth images were stored as 16-bit PNGs and scaled to their respective ranges: [0,80] m for KITTI and [0,250] m for D-Far250, with invalid pixels masked during loss computation.

Multiple architectural combinations were explored to achieve efficient long-range depth estimation, including different encoder depths (ResNet-18 vs. deeper variants), various decoder upsampling strategies (bilinear vs. nearest-neighbor interpolation), and adjustments to multiscale supervision. Final training

was conducted for 1000 epochs with a batch size of 64, using an initial learning rate of 1×10^{-4} . Early stopping was applied with a patience of 15 and a minimum improvement threshold of 0.0001 to prevent overfitting. A ResNet-18 encoder and bilinear upsampling decoder were chosen for the final model to balance accuracy and inference speed.

4.2 | Quantitative Evaluation

To objectively assess the performance of our monocular depth estimation framework, we conduct experiments on both datasets. Both datasets store depth maps as 16-bit PNG images for consistent scaling; however, KITTI provides sparse LiDAR-based ground truth up to 80 m, whereas D-Far250 provides dense synthetic ground truth up to 250 m. Evaluation is performed using the standard metrics defined in Section 3.5.

As shown in Table 2, the proposed model achieves strong performance across both domains. On KITTI, it achieves an AbsRel of 0.0966 and an RMSE of 4.0025 m, with a δ_1 accuracy of 90.85%, validating its reliability in real-world urban driving. On D-Far250, which extends the prediction range to 250 m, the model achieves strong accuracy (AbsRel 0.0655, RMSE 4.90 m, δ_1 94.98%), demonstrating robustness for long-range perception in controlled synthetic environments.

Benchmarking on KITTI: To further validate our framework, we compare it against recent supervised monocular depth estimation methods on the KITTI dataset. Table 3 presents a comparison across commonly used metrics. The proposed method achieves the best RMSE and SqRel scores, while remaining competitive in AbsRel and RMSE(log). Significantly, while

TABLE 2 | Quantitative evaluation of the proposed depth estimation model on KITTI and D-Far250.

Dataset	AbsRel	SqRel	RMSE (m)	RMSElog	δ_1	δ_2	δ_3
KITTI	0.0966	0.0064	4.0025	0.1608	0.9085	0.9759	0.9907
D-Far250	0.0655	0.0026	4.9000	0.1326	0.9498	0.9832	0.9917

Note: SqRel values are reported as implemented in our evaluation script; direct numerical comparison with prior work should consider potential differences in masking and depth scaling.

TABLE 3 | Quantitative comparison of supervised monocular depth estimation methods on the KITTI dataset. The proposed framework is benchmarked against SOTA approaches.

Method	↓ RMSE	↓ RMSE (log)	↓ AbsRel	↓ SqRel	↑ δ_1	↑ δ_2	↑ δ_3
Eigen et al. [1]	6.307	0.282	0.203	1.548	0.702	0.890	0.958
Godard et al. [14]	4.863	0.194	0.115	0.903	0.877	0.959	0.981
Kuznietsov et al. [35]	4.621	0.189	0.113	0.741	0.862	0.960	0.986
Yang et al. [36]	4.442	0.187	0.097	0.734	0.888	0.964	0.990
Zhang et al. [37]	4.137	0.137	0.101	N/A	0.895	0.966	0.991
AdaBins [18]	4.839	0.190	0.107	0.731	0.903	0.964	0.989
Ranftl et al. (DPT/MiDaS) [17]	4.973	0.191	0.110	0.748	0.899	0.962	0.987
ZoeDepth [31]	4.713	0.188	0.101	0.702	0.915	0.968	0.991
Depth Anything [32]	4.650	0.186	0.102	0.698	0.917	0.970	0.993
Our (KITTI)	4.0025	0.1608	0.0966	0.0064	0.910	0.966	0.990

transformer-based models (e.g., AdaBins, DPT, ZoeDepth, Depth Anything) report strong δ accuracies, they typically predict relative depth with scale recovery. In contrast, our approach provides direct 16-bit metric depth, ensuring physically meaningful outputs without additional post-processing.

Although DPT/MiDaS and ZoeDepth are included in Table 3, these transformer-based models are trained using relative depth objectives and post-hoc scale recovery, whereas our network follows direct metric depth training. Due to the substantial differences in their training pipelines and the nonreproducibility of their large-scale data mixtures, we did not retrain these models under identical conditions. Their reported results are therefore used only as broad contextual references rather than strict one-to-one comparisons.

Although some state-of-the-art methods report lower AbsRel values, many of these models rely on highly complex architectures and extensive training pipelines, including multistage optimization and large-scale pretraining. In contrast, the proposed approach aims to achieve a balanced trade-off between accuracy and computational efficiency. The model maintains competitive performance while significantly reducing architectural complexity and training overhead. This design makes the framework particularly suitable for real-time applications and deployment in resource-constrained environments where lightweight and efficient models are preferred over computationally intensive solutions. Importantly, our goal is not to maximize a single error metric in isolation, but to deliver a practically deployable solution that achieves competitive accuracy with substantially lower computational and training overhead, thereby offering a more favorable accuracy–efficiency trade-off than heavier SOTA pipelines.

It is important to emphasize that the objective of the proposed framework is not solely to minimize individual error metrics such as AbsRel, but to provide a balanced solution suitable for real-time cyber-physical mobility systems. Many recent state-of-the-art models achieve lower numerical errors using deep transformer architectures and large-scale pretraining pipelines, which substantially increase computational cost, memory footprint,

and the burden on reproducibility. In contrast, our approach provides (i) direct 16-bit *metric* depth outputs without post-hoc scale alignment, (ii) a lightweight ResNet-based design enabling real-time inference, and (iii) an end-to-end perception pipeline that integrates depth with YOLOv8-based object-wise distance estimation. Therefore, the contribution of this work lies in the accuracy–efficiency–deployability trade-off and the practical distance-aware perception capability, rather than optimizing AbsRel in isolation.

Benchmarking on D-Far250: Since D-Far250 is newly introduced in this study, we perform internal benchmarking across model variants. Table 4 highlights the incremental gains from architectural choices. Moving to ResNet18 with multiscale supervision yields consistent improvements in AbsRel and RMSE. Further integration of detection-guided distance fusion provides the best overall results, with δ_1 accuracy reaching 94.98%. This confirms the effectiveness of our modular design for long-range depth estimation.

Together, these evaluations establish both the competitiveness and the significance of our approach. On KITTI, our method performs on par with or better than state-of-the-art methods while uniquely offering 16-bit metric depth. On D-Far250, we demonstrate scalability to 250 m, a range rarely addressed in existing literature. The combination of real-world benchmarking and synthetic long-range evaluation underscores the novelty of our contributions and their practical relevance for safety-critical applications, such as collision avoidance and autonomous driving.

Complexity Analysis: As shown in Table 5, recent transformer-based methods employ substantially larger model architectures with significantly higher computational complexity. While these models benefit from large-scale pretraining and extensive datasets, they require considerably more parameters and computational resources. In contrast, the proposed ResNet-18-based architecture remains lightweight, requiring only 11.2M parameters and 2.1 GFLOPs, which makes it more suitable for deployment in resource-constrained environments such as edge devices and real-time robotic perception systems. Complexity numbers

TABLE 4 | Benchmarking of proposed framework variants on the D-Far250 dataset. This internal benchmarking highlights improvements from architecture refinements and detection-guided fusion.

Model variant	↓ RMSE (m)	↓ RMSE (log)	↓ AbsRel	↓ SqRel	↑ δ_1	↑ δ_2	↑ δ_3
ResNet18 with DenseNet Decoder	7.523	0.162	0.138	0.0091	0.885	0.955	0.981
ResNet18 with DenseNet-169 Decoder	6.972	0.150	0.121	0.0083	0.902	0.963	0.986
ResNet18 + Multiscale Supervision	4.9000	0.1326	0.0655	0.0026	0.9498	0.9832	0.9917

TABLE 5 | Model complexity comparison with recent transformer-based depth estimation methods. Transformer models typically use significantly larger architectures and are often trained on substantially larger datasets. In contrast, the proposed ResNet-18-based model maintains competitive performance while remaining lightweight and computationally efficient.

Method	Backbone type	Params (M)	GFLOPs
Proposed (Ours)	ResNet-18 (CNN)	11.2	2.1
Depth Anything V2	Vision Transformer	~335	~58
ZoeDepth	Transformer (DPT backbone)	~345	~60
DPT-Hybrid (MiDaS)	CNN + Transformer	~123	~35

for transformer baselines are taken from their official papers or repositories [17, 31, 32].

4.3 | Qualitative Results

To complement the quantitative analysis, we present qualitative examples of the predicted depth maps compared with the ground truth and the input RGB images. These samples are drawn from the KITTI validation split and illustrate the model's capability to capture geometric structures and depth gradients across diverse urban driving scenes. The predicted depth maps are visualized using the perceptually uniform viridis colormap after normalization to the range [0, 80] m. Brighter tones correspond to objects at greater distances, while darker tones denote objects closer to the viewer. The 16-bit output maps are normalized and rescaled for visualization while preserving the original scale.

Figure 2 effectively shows recovered object boundaries, road surfaces, and varying depth layers with high spatial coherence. It is essential to note that some parts of the predicted depth map appear darker, corresponding to smaller depth values under the applied colormap and visualization normalization. As we are dealing with 16-bit images, some regions may appear darker. The predicted maps are well aligned with the sparse LiDAR ground truth, especially in the foreground and mid-range areas. Performance may degrade in low-texture regions and at object edges, a common limitation in monocular depth estimation.

To visually validate the model's prediction quality, we compare RGB images from the D-Far250 dataset with their corresponding

predicted depth maps and 16-bit ground-truth depth maps. Figure 3 shows three representative examples covering long-range urban scenes with depth values extending up to 250 m.

The predicted depth maps effectively capture road curvature, object shapes, and horizon alignment, even in challenging low-texture regions such as sky boundaries and distant trees. The use of 16-bit encoding preserves fine-grained depth transitions, which are evident in both mid-range and far-field areas. Despite the large spatial scale, the network demonstrates consistent predictions across frames.

It is important to note that the predicted depth maps may appear visually darker than typical depth visualizations. This is not an indication of poor performance but rather a result of the extended depth range (up to 250 m) and high 16-bit pixel precision used in both prediction and ground truth. Since closer objects occupy fewer pixels and most distant surfaces dominate the frame, the color mapping is skewed toward lower-intensity tones. Nonetheless, the predicted maps accurately encode metric depth values across the full spatial range, and this high-resolution fidelity is preserved despite the perceptual compression in the color gradient.

4.4 | Object Detection and Distance Estimation

In this stage, we perform object detection on monocular RGB images using a fine-tuned YOLOv8 model, followed by distance inference for each detected object utilizing the predicted depth maps. The YOLOv8 model was fine-tuned on both the KITTI

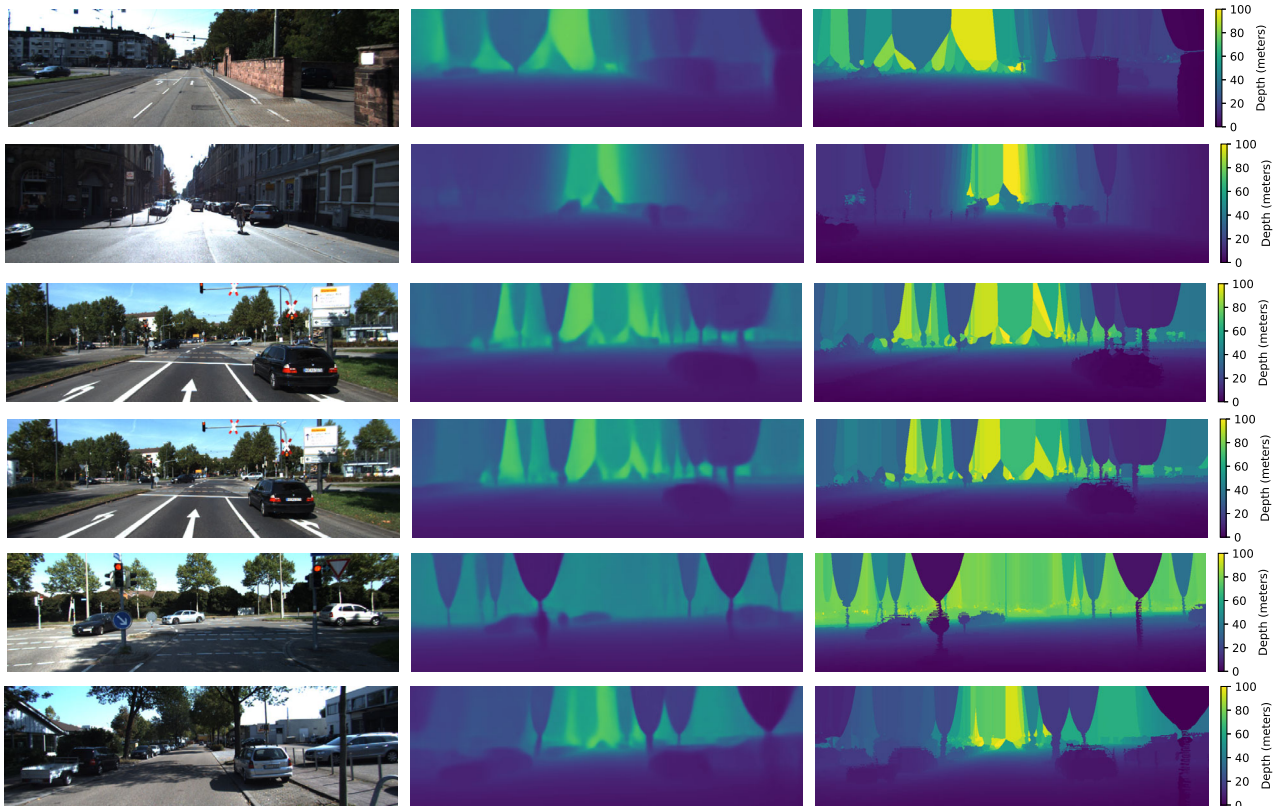


FIGURE 2 | Left: RGB input; middle: predicted depth map; right: ground truth depth. Our method accurately captures object depth and scene geometry. Darker regions correspond to nearer objects while brighter tones represent farther surfaces, following the applied viridis depth visualization.

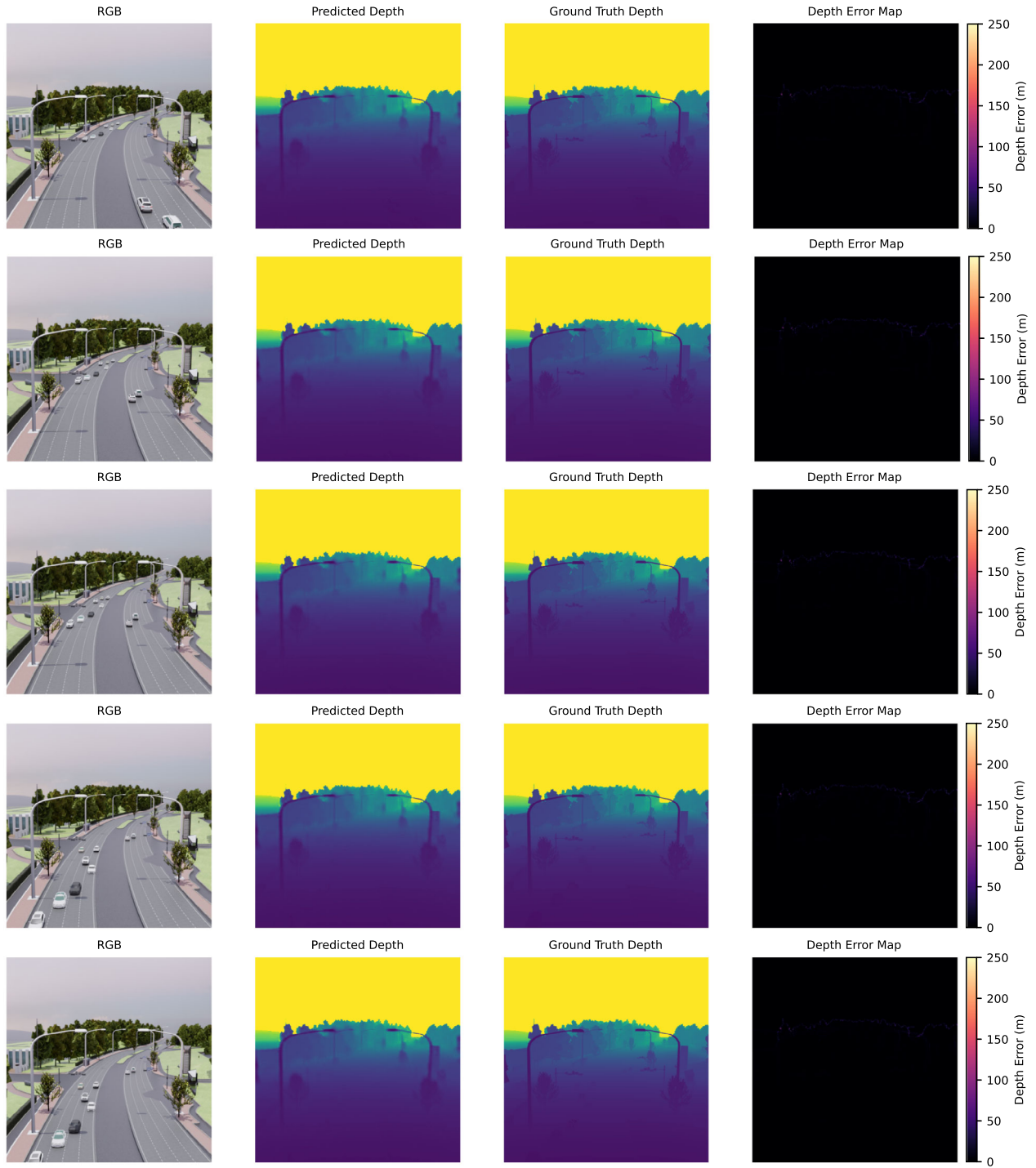


FIGURE 3 | Qualitative results on D-Far250 dataset. Each row, from left to right, shows the RGB image, the predicted depth map, and the ground-truth depth map. The model produces spatially consistent and geometrically accurate depth estimates even at ranges of up to 250 m. Yellow areas represent maximum depth values.

dataset (real-world short-range scenes) and the D-Far250 dataset (synthetic long-range scenes) using the official Ultralytics implementation. For each image, object detection yields bounding boxes $B = \{b_1, b_2, \dots, b_n\}$, class labels $C = \{c_1, c_2, \dots, c_n\}$, and confidence scores.

Let the predicted depth map be $D \in \mathbb{R}^{H \times W}$, where each pixel stores depth in meters. For a given bounding box $b_i = (x_{\min}, y_{\min},$

$x_{\max}, y_{\max})$, we extract the set of depth pixels inside the detected region

$$R_i = \{D(x, y) \mid x_{\min} \leq x < x_{\max}, y_{\min} \leq y < y_{\max}\} \quad (24)$$

The estimated object distance $\hat{d}_i$ is computed using the median depth value within the bounding region

$$\hat{d}_i = \text{median}_{(x,y) \in R_i} D(x, y) \quad (25)$$

The median statistic is preferred because it is robust to outliers and depth noise near object boundaries, producing more stable distance estimates than single-pixel sampling. Although median depth pooling is robust to isolated outliers, certain driving scenarios may introduce estimation bias. For example, when one car partially occludes another, or when a distant vehicle occupies only a small portion of the bounding box, background pixels (road or sky) may dominate the depth statistics. Such cases were occasionally observed in long-range scenes of the D-Far250 dataset, where very distant cars or partially occluded vehicles introduce larger depth variance within the bounding region.

To evaluate accuracy, the ground truth distance d_i (from KITTI 3D labels) is compared to $\hat{d}_i$, and the absolute error e_i is

$$e_i = |d_i - \hat{d}_i| \quad (26)$$

To further analyze the behavior of the proposed distance estimation strategy, we perform a depth-range bin evaluation on the KITTI dataset, summarized in Table 6. Visual results are shown in Figure 4 as an example, where green bounding boxes indicate detected objects and their estimated distances. Instead of listing individual detections, objects are grouped into predefined distance intervals and evaluated using standard error metrics. For each distance bin, we report the number of matched detections, along with the MAE and RMSE computed from the region-median depth values within the detected bounding boxes. This range-wise analysis enables systematic examination of how distance estimation accuracy varies across near- and mid-range objects in real-world driving scenes. Reporting the results in distance bins avoids scale bias that could arise from aggregating all objects into a single global MAE or RMSE, particularly because the evaluated objects differ significantly in size, distance, and LiDAR sampling density. Therefore, Table 6 emphasizes consistent distance estimation behavior across different depth ranges rather than relying solely on a single aggregated error metric.

As illustrated in the qualitative examples, the model shows strong agreement with ground truth for many near- and mid-range objects. However, estimation error increases with distance due to reduced pixel coverage and weaker monocular cues, consistent with the range-bin results reported in Table 6. Smaller

TABLE 6 | Depth-range bin evaluation for object-wise distance estimation on KITTI. Errors are computed over matched detections using region-median depth inside each bounding box.

Distance bin, m	#Objects	MAE, m	RMSE, m	GT range, m
0–10	31	0.68	1.08	2.33–9.08
10–20	45	0.78	1.05	10.06–18.09
20–30	29	3.78	5.72	21.35–24.33
30–40	21	2.81	4.37	31.58–38.83
40–50	36	3.11	4.72	42.12–47.94

Note: The GT ranges reflect matched detections available in the selected KITTI validation scenes; bins beyond 50 m are not reported here because the sampled scenes in this analysis contain no matched objects in those ranges.

objects, such as pedestrians and thin structures, remain challenging because bounding-box regions contain fewer reliable depth pixels, whereas median-based aggregation improves robustness to boundary noise and outliers.

It is also noteworthy that smaller objects, such as pedestrians and narrow vertical structures (e.g., streetlights), are predicted with consistent accuracy. The error margin for pedestrian detection hovers below 1 m, which is critical for safety-centric applications such as autonomous navigation or ADAS.

While the KITTI evaluation focuses on near- and mid-range objects typically encountered in urban driving environments, the D-Far250 dataset is used to further analyze long-range distance estimation performance up to 250 m.

Table 7 presents a depth-range bin analysis of object-wise distance estimation on the D-Far250 dataset. Instead of listing individual detections, objects are grouped into distance intervals to systematically evaluate the distribution of estimation errors across ranges up to 250 m. For each bin, we report the number of matched detections, along with the MAE and RMSE computed from the region-median depth values within the detected bounding boxes. Range-wise error analysis is reported on the D-Far250 dataset because it contains objects distributed across significantly larger distances (up to 250 m), enabling a meaningful evaluation of long-range depth estimation behavior. In contrast, KITTI primarily contains near- to mid-range objects and therefore provides limited coverage for the analysis of far-distance error distributions.

These evaluations confirm that the proposed model not only generates visually accurate dense depth maps but also reliably estimates true distances for critical objects within a 250 m visual range, thereby validating the suitability of the D-Far250 dataset and our model pipeline for long-range perception tasks.

Range-wise error analysis: The results indicate that estimation accuracy varies with object distance, which is expected in monocular depth estimation due to reduced pixel resolution and weaker geometric cues for distant objects. Near and mid-range objects show relatively low errors, while far-range objects exhibit slightly larger deviations as depth uncertainty increases with distance. Nevertheless, the median-based estimation strategy provides stable distance predictions across all evaluated ranges, supporting reliable object-level distance estimation up to 250 m.

The object-wise distance-estimation results on the D-Far250 dataset demonstrate the proposed model's strong generalization across both near- and long-range targets (Figure 5). Despite the dataset extending up to 250 m, the predicted distances closely match the ground truth, despite the dataset extending up to 250 m, the predicted distances closely match the ground truth, typically within 5 m, with occasional higher deviations depending on object size and distance. The model maintains robustness across varying scales and depth ranges, with particularly low deviations for objects beyond 120 m, a challenging regime for monocular systems. These findings confirm the effectiveness of our ResNet-based multiscale decoder architecture, trained



FIGURE 4 | Qualitative results showing detected objects and estimated distances from predicted depth maps. Green bounding boxes denote detected objects with overlaid distance estimates. Range-wise error statistics for these KITTI scenes are summarized in Table 6.

TABLE 7 | Depth-range bin evaluation for object-wise distance estimation on D-Far250. Errors are computed over IoU-matched car detections using region-median depth inside each box.

Distance bin (m)	#Objects	MAE, m	RMSE, m	GT range, m
0–50	13	2.72	3.21	26.25–37.54
50–100	4	1.45	1.65	51.53–56.38
100–150	18	3.28	3.89	115.54–135.93
150–200	2	0.61	0.80	157.44–181.86
200–250	4	2.95	3.29	221.03–238.64

Note: The GT ranges reflect matched detections available in the validation split; bins may not cover boundaries.

with consistent loss formulations across datasets, and further validate the utility of depth maps for high-precision object localization in extended-view scenarios.

4.5 | Detection and Runtime Evaluation

In addition to standard depth estimation metrics (AbsRel, RMSE, LogRMSE, and δ thresholds), we provide complementary evaluation results that strengthen the comprehensiveness of our study.

These include object detection metrics (mAP, mAR) obtained by coupling our framework with YOLOv8, as well as real-time performance indicators (FPS and latency) measured on both GPU and CPU platforms. Together, these metrics provide a broader view of the proposed system’s performance in realistic deployment scenarios where both accuracy and efficiency are critical.

4.5.1 | Object Detection Metrics

To validate the utility of the proposed framework in downstream perception tasks, we benchmarked detection accuracy on the D-Far250 validation set. Table 8 reports Precision, Recall, and mean Average Precision (mAP) at IoU thresholds of 0.5 and 0.5–0.95. The final epoch (85) achieved the highest mAP@50 score of **0.8670**, while the best-performing epoch (83) yielded the best Recall (0.8396) and mAP@50–95 (0.4008). These results indicate that our system produces reliable detection outputs while maintaining strong depth estimation fidelity.

4.5.2 | Runtime Benchmarking

Real-time performance is essential for deploying monocular depth estimation in autonomous and IoT-based systems. We benchmarked inference speed on both an NVIDIA GeForce RTX 4070 GPU and an Intel i9 CPU. As shown in Table 9, the framework achieves 221.96 FPS on the GPU with an average

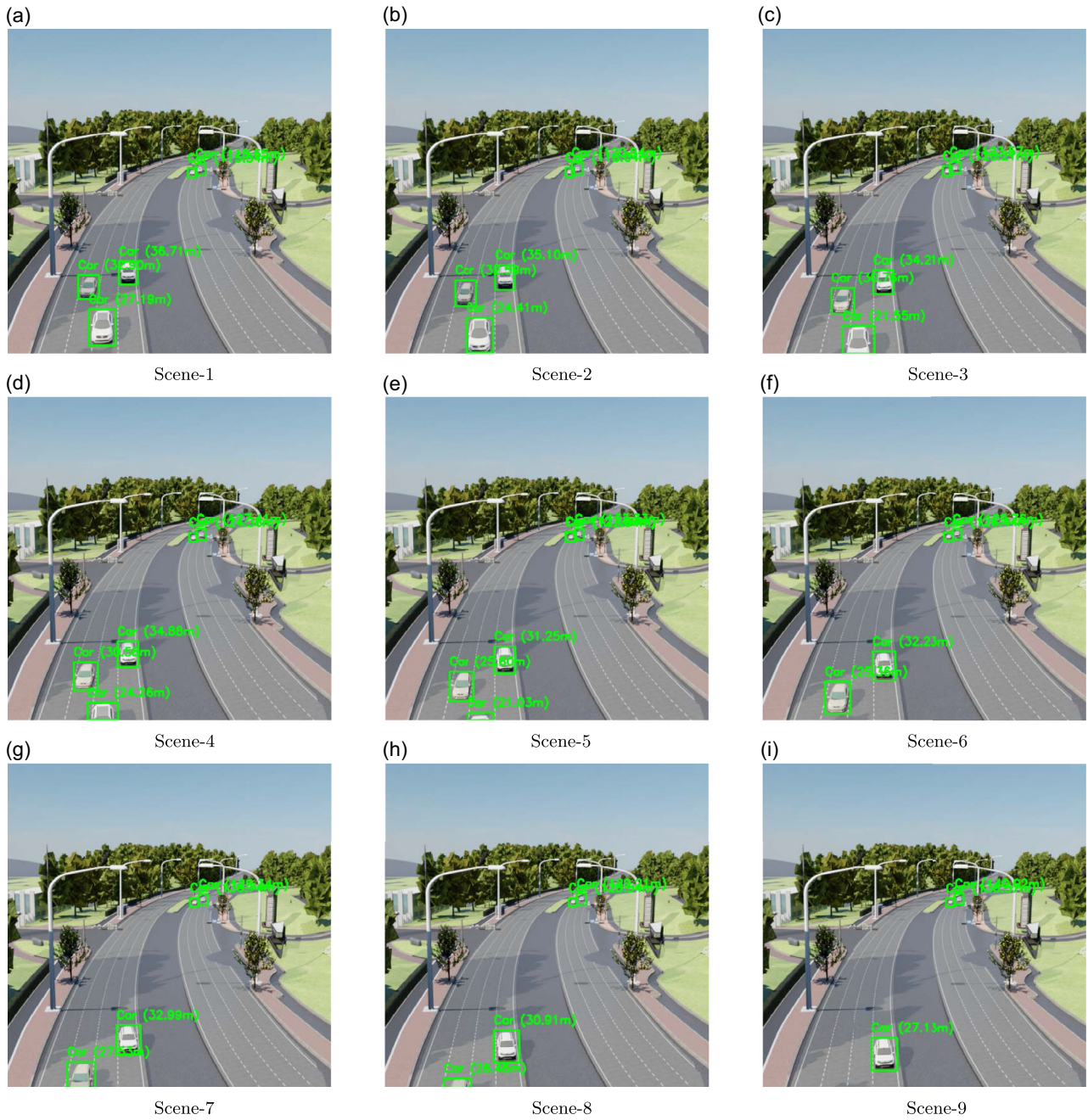


FIGURE 5 | Qualitative results showing detected objects and estimated distances from predicted depth maps on the D-Far250 dataset. Bounding boxes indicate detected vehicles, with predicted distance values overlaid. The corresponding range-wise error summary is reported in Table 7.

TABLE 8 | Object detection performance of YOLOv8 on the D-Far250 validation set, reported at the final and best-performing epochs.

Epoch	Precision	Recall	mAP@50	mAP@50-95
Final (85)	0.8185	0.8299	0.8670	0.3965
Best (83)	0.8129	0.8396	0.8624	0.4008

latency of 4.51 ms per frame, comfortably exceeding the standard 30 FPS real-time requirement. In contrast, the CPU reaches 10.76 FPS with a latency of 92.97 ms, which, while below real-time thresholds, is included to demonstrate portability rather than to serve as the target deployment configuration.

It is important to note that contemporary IoT and edge-AI platforms increasingly rely on hardware accelerators such as embedded GPUs, NPUs, and AI coprocessors (e.g., NVIDIA Jetson Orin, Google Edge TPU), which offer performance characteristics much closer to our GPU results than to CPU-only execution. Therefore, the real-time capability claim is based on accelerator-enabled inference, consistent with modern IoT deployment practices, while CPU throughput serves as a fallback for nontime-critical scenarios.

Beyond depth-specific metrics, we further extend the evaluation by incorporating detection-driven performance and runtime efficiency, providing a holistic view of the framework's accuracy and deployability (Figure 6). Overall, the extended evaluation

TABLE 9 | Runtime benchmarking results of the proposed framework on GPU and CPU platforms. FPS and latency values are averaged over the test set.

Device	Average latency, ms/frame	Throughput, FPS
GPU (RTX 4070)	4.51	221.96
CPU (Intel i9)	92.97	10.76

confirms that our method not only achieves competitive depth accuracy but also supports reliable detection performance and real-time execution. This combination underscores the practical significance of the proposed system for safety-critical domains such as intelligent transportation and autonomous navigation.

5 | Discussion

The comprehensive evaluation of the proposed depth estimation framework on KITTI and D-Far250 demonstrates both its potential and its limitations, offering critical insights into the behavior of monocular systems across different domains and ranges. It is important to contextualize the reported results across the KITTI and D-Far250 datasets to avoid overstating performance. While D-Far250 yields numerically lower errors (e.g., RMSE $\approx$ 4.90 m), these values are partly influenced by the dataset's dense synthetic ground truth and by scaling. Object-level analysis reveals that errors increase progressively beyond 125 m, reflecting the inherent difficulty of long-range monocular perception, in which distant objects occupy few pixels. Thus, D-Far250 results should be interpreted as evidence of scalability to extended ranges rather than direct proof of generalization to real-world conditions. KITTI, by contrast, validates the framework's ability to generalize in practical driving scenarios up to 80 m. Similarly, while our

framework achieves over 200 FPS on an NVIDIA RTX 4070 GPU, this does not directly translate to edge devices, where further optimization will be required. Finally, although D-Far250 introduces greater variability in object types and depth ranges, it does not fully capture real-world complexities such as adverse weather or lighting, which remain critical directions for future evaluation. Although experiments were conducted on an NVIDIA RTX 4070 GPU to ensure reproducible benchmarking and fair comparison with existing depth-estimation studies, the proposed architecture is lightweight and computationally efficient, making it suitable for deployment on resource-constrained edge platforms such as embedded GPUs or edge AI accelerators.

Performance Across Ranges: The strong performance on KITTI confirms the model's reliability in real-world urban driving up to 80 m, while the D-Far250 dataset extends this to 250 m under synthetic conditions. The near-equivalence of AbsRel and δ -accuracy across the two benchmarks suggests that the architecture generalizes well to long-range settings. However, the stronger performance on D-Far250 compared to KITTI also indicates a range gap: synthetic environments provide idealized conditions with denser annotations and fewer degradations (e.g., atmospheric scattering, low-light noise), which partially explains the near-perfect scores. This highlights the challenge of directly transferring synthetic robustness to real-world deployment.

Error Behavior at Long Distances: Object-level analysis reveals that errors increase progressively with distance, particularly beyond 125 m. This is expected, as very distant objects occupy only a few pixels, limiting spatial cues, and are more affected by depth compression in 16-bit scaling. While errors increase at long range due to reduced pixel coverage, the range-bin analysis indicates that median-based estimates remain stable in most evaluated cases, with larger deviations occurring as objects occupy only a few pixels and depth uncertainty

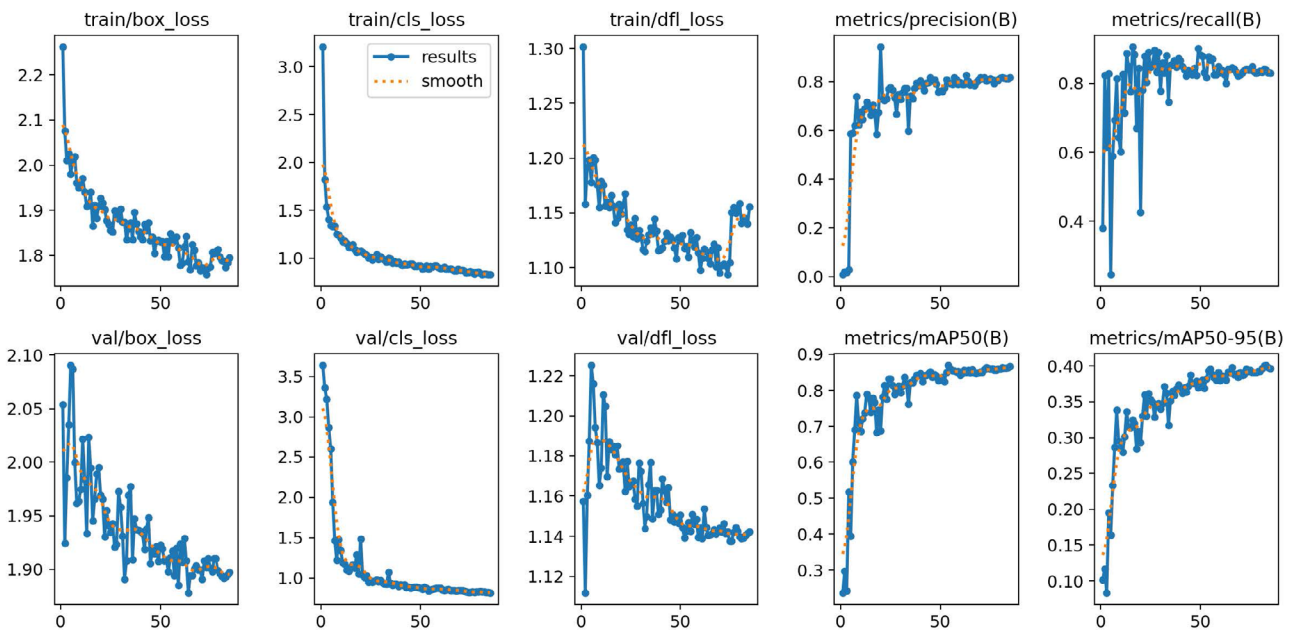


FIGURE 6 | Training and validation curves showing the evolution of losses (box, classification, and focal) alongside detection metrics (precision, recall, and mAP). These results highlight both stable convergence and strong performance in depth estimation guided by detection.

increases. In real-world scenarios, weather conditions such as fog, rain, or glare would likely exacerbate these errors.

Environmental Factors: The role of lighting and weather cannot be overstated. Monocular methods rely heavily on visible texture and contrast, making them sensitive to shadows, nighttime scenes, and adverse weather. While D-Far250 includes some variation, it lacks the full complexity of real-world atmospheric effects. Future extensions must explicitly evaluate robustness under challenging illumination and weather to ensure safe deployment.

Significance for Deployment: Despite these limitations, the integration of YOLOv8 for detection-guided distance inference strengthens the system's practical utility. Accurate mid-range and reasonable long-range depth predictions, combined with real-time inference speed, position the framework as a scalable, LiDAR-free alternative for applications such as autonomous driving, unmanned aerial vehicle (UAV) navigation, and outdoor robotics. The framework's ability to maintain spatial coherence and avoid catastrophic long-range error escalation confirms its readiness for further validation in real-world conditions.

In summary, this work introduces a monocular depth estimation framework that extends the fidelity of metric prediction to the underexplored 250 m range while maintaining competitive performance on the KITTI benchmark. The proposed D-Far250 dataset demonstrates the framework's scalability for long-range perception under controlled conditions; however, these synthetic results should not be interpreted as direct evidence of generalization in real-world settings. Dataset validation was performed by matching KITTI camera intrinsics, horizon geometry, and depth-pixel slope characteristics to ensure consistency with real-world perspective behavior. No domain-adaptation techniques were applied in the current study, as D-Far250 was primarily designed to evaluate long-range scalability rather than synthetic-to-real transfer. Consequently, performance on D-Far250 should be interpreted as evidence of scalability under controlled conditions rather than definitive proof of real-world reliability. Future work will therefore focus on integrating domain adaptation strategies, cross-dataset validation, and robustness evaluation under real-world conditions such as weather variations, sensor noise, and illumination changes to further strengthen deployment reliability. Object-level evaluation further shows increased errors beyond 125m, reflecting the inherent challenges of monocular inference at extreme distances. Additionally, although GPU benchmarking confirms over 200 FPS on an NVIDIA RTX 4070, real-time deployment on edge devices will require further optimization. Collectively, these findings highlight both the strengths and limitations of the proposed approach. Future work will focus on enhancing the realism and diversity of D-Far250 by incorporating adverse weather and illumination conditions, and on advancing synthetic-to-real transfer strategies to support robust deployment in autonomous driving, UAV navigation, and other safety-critical domains.

6 | Conclusion and Future Work

This study presents a real-time monocular depth estimation framework that generates 16-bit metric depth maps up to 250 m using a multiscale supervised encoder-decoder architecture based

on ResNet18. Unlike conventional 8-10-bit or short-range depth approaches, the proposed method preserves fine spatial detail across both near and far distances, enabling reliable long-range perception. Experimental results demonstrate strong performance on the KITTI benchmark (AbsRel 0.0966, $\delta_1 = 90.85\%$) and further confirm scalability through the proposed D-Far250 dataset (AbsRel 0.0655, $\delta_1 = 94.98\%$). By integrating the dense depth pipeline with YOLOv8, the framework enables detection-guided distance estimation, linking dense monocular depth prediction to practical metric distance inference in real-world applications. The system achieves over 200 FPS on an RTX 4070 GPU, highlighting its suitability for latency-sensitive IoT scenarios, including intelligent transportation systems, UAV monitoring, and smart surveillance. Overall, the results demonstrate that high-resolution 16-bit depth modeling and multiscale supervision can significantly enhance long-range monocular perception while maintaining real-time performance.

Despite these contributions, several limitations remain. The evaluation on the D-Far250 dataset relies on synthetic data, which may not fully represent real-world environmental variability or sensor noise. In addition, the current distance estimation strategy, based on median depth pooling within detection bounding boxes, may be less reliable under heavy occlusion or when distant objects occupy only a few pixels. The current implementation also emphasizes GPU-based benchmarking, and further optimization is required for efficient deployment on resource-constrained edge devices. Future research will therefore focus on reducing the synthetic-to-real domain gap through domain adaptation techniques, developing instance-aware depth aggregation methods to improve object-level distance estimation, and validating the framework on additional real-world datasets and embedded edge-computing platforms. These directions aim to enhance robustness and support the scalable deployment of long-range monocular perception systems in next-generation IoT-enabled mobility environments.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The D-Far250 dataset used in this study cannot be fully released due to licensing and distribution restrictions associated with the rendering assets. A reduced 5%–10% benchmark subset will be made publicly available once permissions are finalized. Additional data may be obtained from the corresponding author upon reasonable request.

Declaration of Generative AI Use

During the preparation of this work, the author(s) used ChatGPT and Grammarly to enhance the clarity, coherence, and grammar of the manuscript. Following the use of these tools, the author(s) carefully reviewed and edited the content to ensure accuracy and originality, and take full responsibility for the final content of the publication.

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